Complex arithmetic for signals and systems

Operation	*Formula*
Rectangular to Polar Conversion	$z = x + jy = re^{j\theta}$ where $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$
Polar to Rectangular Conversion	$z = re^{j\theta} = r\,[\cos(\theta) + j\sin(\theta)] = x + jy$ where $r = \cos(\theta)$ and $y = r\sin(\theta)$
Add: $z_3 = z_1 + z_2$	$(x_1 + x_2) + j\,(y_1 + y_2)$
Subtract: $z_3 = z_1 - z_2$	$(x_1 - x_2) + j\,(y_1 - y_2)$
Multiply: $z_3 = z_1 z_2$ (polar form)	$(x_1 x_2 - y_1 y_2) + j\,(x_1 y_2 + y_1 x_2)$ $r_1 r_2\, e^{j(\theta_1 + \theta_2)}$
Divide: $z_3 = z_1 / z_2$ (polar form)	$\dfrac{(x_1 x_2 - y_1 y_2) - j\,(x_1 y_2 - y_1 x_2)}{x_2^2 + y_2^2}$ $\dfrac{r_1}{r_2}\, e^{j(\theta_1 - \theta_2)}$

Trigonometry and Euler's formulas

Relationship	*Relationship*
$\sin u = \cos(u - \pi/2)$	$\cos u = \sin(u + \pi/2)$
$\cos(-u) = \cos u$	$\sin(-u) = -\sin(u)$
$\sin^2 u + \cos^2 u = 1$	$\cos^2 u = \frac{1}{2}(1 + \cos 2u)$
$\sin^2 u = \frac{1}{2}(1 - \cos 2u)$	$\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$
$\sin(u \pm v) = \sin u \cos v \pm \cos u \sin v$	$\cos u \cos v = \frac{1}{2}[\cos(u - v) + \cos(u + v)]$
$\sin u \sin v = \frac{1}{2}[\cos(u - v) + \cos(u + v)]$	$\sin u \cos v = \frac{1}{2}[\sin(u - v) + \sin(u + v)]$
$\cos u = \frac{1}{2}\left[e^{ju} + e^{-ju}\right]$	$\sin u = \frac{1}{2j}\left[e^{ju} - e^{-ju}\right]$
$e^{ju} = \cos u + j\sin u$	

Geometric series

Name	*Sum*	*Condition*
Finite on $[N_1, N_2]$	$\sum_{n=N_1}^{N_2} a^n = \frac{a^{N_1} - a^{N_2+1}}{1-a}$	none
Finite on $[0, N-1]$	$\sum_{k=0}^{N} a^k = \frac{1-a^{N+1}}{1-a}$	none
Infinite	$\sum_{n=0}^{\infty} a^k = \frac{1}{1-a}$	$\lvert a \rvert < 1$

Recognizing signal properties and classifications

Property	*Continuous*	*Discrete*
Energy	$E_x = \int_{-\infty}^{\infty} \lvert x(t) \rvert^2 dt < \infty$	$E_x = \sum_{n=-\infty}^{\infty} \lvert x[n] \rvert^2$
Power	$P_x = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \lvert x(t) \rvert^2 dt < \infty$	$P_x = \lim_{N\to\infty} \frac{1}{2N} \sum_{n=-\infty}^{\infty} \lvert x[n] \rvert^2 < \infty$
Periodic	$x(t - T_0) = x(t),\ T_0 = \text{period}$	$x[n - N_0] = x[n],\ N_0 = \text{period}$
Even	$x(-t) = x(t)$	$x[-n] = x[n]$
Odd	$x(-t) = -x(t)$	$x[-n] = -x[n]$

Name	*Continuous*	*Discrete*
Impulse	$\delta(t) = 0,\ t \neq 0$ $\int_{-\infty}^{\infty} \delta(t)dt = 1$	$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$
Step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$	$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$
Rectangle Pulse	$\Pi\left(\frac{t}{\tau}\right) = \begin{cases} 1, & t \leq \lvert\tau/2\rvert \\ 0, & t > \lvert\tau\rvert \end{cases}$	$u[n] - u[n-M]$
sinc( ) and aliased sinc( )	$\text{sinc}(t) = \frac{\sin \pi t}{\pi t}$	$\frac{\sin(N\pi x)}{\sin(\pi x)}$, aliased sinc()
Sinusoid	$A\cos(2\pi f_0 t + \phi)$	$A\cos(\hat{\omega}_0 n + \phi)$, $\hat{\omega}$ is mod 2π

Recognizing system properties and classifications

Property	*Continuous*	*Discrete*
Input/Output	$x(t) \rightarrow \boxed{h(t)} \rightarrow y(t)$	$x[n] \rightarrow \boxed{h[n]} \rightarrow y[n]$
Time	$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t-\lambda)d\lambda$	$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$
Frequency	$Y(f) = X(f)H(f)$	$Y(e^{j\hat{\omega}}) = X(e^{j\hat{\omega}})H(e^{j\hat{\omega}})$
s/z-Domain	$Y(s) = X(s)H(s)$	$Y(z) = X(z)H(z)$
Cascading	$\rightarrow \boxed{h_1(t)} \rightarrow \boxed{h_2(t)} \rightarrow$	$\rightarrow \boxed{h_1[n]} \rightarrow \boxed{h_2[n]} \rightarrow$
Time	$h(t) = h_1(t) * h_2(t)$	$h[n] = h_1[n] * h_2[n]$
Frequency	$H(f) = H_1(f) \cdot H_2(f)$	$H(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}}) \cdot H_2(e^{j\hat{\omega}})$
s/z-Domain	$H(s) = H_1(s) \cdot H_2(s)$	$H(z) = H_1(z) \cdot H_2(z)$
LCCDE	$\sum_{k=1}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$	$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$
Frequency	$H(f) = \frac{Y(f)}{X(f)} = \sum_{k=1}^{M} b_k (j2\pi f)^k \Big/ \sum_{k=0}^{N} a_k (j2\pi f)^k$	$H(e^{j\hat{\omega}}) = \frac{Y(e^{j\hat{\omega}})}{X(e^{j\hat{\omega}})} = \sum_{k=0}^{M} b_k e^{-j\hat{\omega}k} \Big/ \sum_{k=0}^{N} a_k e^{-j\hat{\omega}k}$
s/z-Domain	$H(s) = Y(s)/X(s) = \sum_{k=0}^{M} b_k s^k \Big/ \sum_{k=0}^{N} a_k s^k$	$H(z) = Y(z)/X(z) = \sum_{k=0}^{M} b_k z^{-k} \Big/ \sum_{k=0}^{N} a_k z^{-k}$
Impulse Resp	$(h)t = \mathcal{L}^{-1}\{(H)s,\}$ partial fractions	$h[n] = \mathcal{Z}^{-1}\{H(z)\}$, partial fractions
Arbitrary Resp	$(y)t = \mathcal{L}^{-1}\{(X)s(H)s,\}$ partial fractions	$[y\ n] = \mathcal{Z}^{-1}\{X(z)H(z)\}$, partial fractions
Sinusoid SS	$x(t) = A\cos(2\pi f_0 t + \phi)$ $y(t) = A\lvert H(f_0)\rvert\cos(2\pi f_0 t + \phi + \angle H(f_0))$	$x[n] = A\cos(\hat{\omega}_0 n + \phi)$ $y[n] = A\lvert H(e^{j\hat{\omega}_0})\rvert\cos(\hat{\omega}_0 n + \phi + \angle H(e^{j\hat{\omega}_0}))$
Stability		
Time	$\int_{-\infty}^{\infty} \lvert h(t)\rvert dt < \infty$	$\sum_{n=-\infty}^{\infty} \lvert h[n]\rvert < \infty$
s/z-Domain	$j\omega$-axis $\subset$ Region of convergence	unit circle $\subset$ Region of convergence
s/z-Causal	poles in left-half plane	poles inside unit circle

Using Fourier transforms for continuous-time signals

$$\text{Frequency variable } f \begin{cases} X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt \\ x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft}\, dt \end{cases}$$

$$\text{Frequency variable } \omega \begin{cases} X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt \\ x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega \end{cases}$$

Property / Pair	***Signal***	***FT in*** f	***FT in*** ω
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(f) + bX_2(f)$	$aX_1(\omega) + bX_2(\omega)$
Time delay	$(xt - t_0)$	$(X) f e^{-j2\pi f t_0}$	$(X)\, \omega e^{-j\omega t_0}$
Frequency Translation	$x(t) e^{j2\pi f_0 t}$	$X(f - f_0)$	$X(\omega - \omega_0)$
Convolution	$x_1(t) * x_2(t)$	$X_1(f) \cdot X_2(f)$	$X_1(\omega) \cdot X_2(\omega)$
Multiplication	$x_1(t) \cdot x_2(t)$	$X_1(f) * X_2(f)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Parseval' s Theorem	$\int_{-\infty}^{\infty} \lvert x(t) \rvert^2 dt$	$\int_{-\infty}^{\infty} \lvert X(f) \rvert^2 df$	$\frac{1}{2\pi} \int_{-\infty}^{\infty} \lvert X(\omega) \rvert^2 d\omega$
Rectangle	$\Pi\left(\frac{t}{\tau}\right)$	$\tau\, \mathrm{sinc}(f\tau)$	$\tau\, \mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
sinc( )	$2W\ \mathrm{sinc}(2Wt)$	$\Pi\left(\frac{f}{2W}\right)$	$\Pi\left(\frac{\omega}{4\pi W}\right)$
Triangle	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau\, \mathrm{sinc}^2(f\tau)$	$\tau\, \mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
Exponential	$e^{-at}(u), t \quad a>0$	$\frac{1}{a + j2\pi f}$	$\frac{1}{a + j\omega}$
Impulse	$A\delta(t)$	A	A
Constant	A	$A\delta(f)$	$2\pi A\delta(\omega)$
Complex exponential	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$	$2\pi\delta(\omega - \omega_0)$

Assume that $x_1(t)$ and $x_2(t)$ have FTs $X_1(f)$ and $X_2(f)$ respectively.

Applying Fourier transform to discrete-time signals

Property / Pair	*Signal*	*Transform*
Linearity	$ax_1[n]+bx_2[n]$	$aX_2(e^{j\hat{\omega}})+bX_2(e^{j\hat{\omega}})$
Time shift	$x[n-n_0]$	$e^{-j\hat{\omega}n_0}X(e^{j\hat{\omega}})$
Frequency shift	$e^{j\hat{\omega}_0 n}[x]n$	$X(e^{j(\hat{\omega}-\hat{\omega}_1)})$
Convolution	$x[n]*h[n]$	$X(e^{j\hat{\omega}})H(e^{j\hat{\omega}})$
Multiplication	$x[n]w[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\theta})W(e^{j(\hat{\omega}-\theta)})d\theta$
Parseval's	$\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2$	$\frac{1}{2\pi}\int_{-\pi}^{\pi}\lvert X(e^{j\hat{\omega}})\rvert^2 d\hat{\omega}$
Impulse	$\delta[n]$	1
Exponential	$a^n[u]n\ (\lvert a\rvert<1)$	$\frac{1}{1-ae^{-j\hat{\omega}}}$
Rectangle	$u[n]-u[n-M]$	$\frac{\sin(\hat{\omega}M/2)}{\sin(\hat{\omega}/2)}\cdot e^{-j\hat{\omega}(M-1)/2}$
Rectangle in Frequency	$\frac{\sin\hat{\omega}_c n}{\pi n}$	$\Pi\left(\frac{\hat{\omega}}{2\hat{\omega}_c}\right)$
Constant	1	$\pi\delta(\hat{\omega}), -\pi<\hat{\omega}\le\pi$
Complex exponential	$e^{j\hat{\omega}_0 n}$	$2\pi\delta(\hat{\omega}-\hat{\omega}_0), -\pi<\hat{\omega}\le\pi$
Assume that $x_1[n]$ and $x_2[n]$ have DT FTs $X_1(e^{j\hat{\omega}})$ and $X_1(e^{j\hat{\omega}})$ respectively.		

$$X\left(e^{j\hat{\omega}}\right)=\sum_{n=-\infty}^{\infty}x[n]e^{j\hat{\omega}n} \text{ and } x[n]=\frac{1}{2\pi}\int_0^{2\pi}X\left(e^{j\hat{\omega}}\right)e^{j\hat{\omega}t}\,d\hat{\omega}$$

Using the Laplace transform in the s-domain

$$X(s) = \int_0^{\infty} x(t) e^{jst}\, dt$$

Property / Pair	$x(t)$	$X(s)$	***ROC***
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	at least $R_{x_1} \cap R_{x_2}$
Time Shift	$x(t - t_0)u(t - t_0)$	$e^{-st_0}X(s)$	R_x
Differentiation	$\frac{d^n x(t)}{dt^n}$	$sX(s) - s^{n-1}x(0^-) - \cdots - x^{(n-1)}(0^-)$	R_x
Integration	$\int_{0^-}^{\infty} x(\lambda)\, d\lambda$	$\frac{X(s)}{s}$	$R_x \cap (\mathrm{Re}(s) > 0)$
Convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	at least $R_{x_1} \cap R_{x_2}$
Final value	$x(\infty) =$	$\lim_{s \to 0} sX(s)$	if limit exists
Impulse	$\delta(t)$	1	all s
Exponential	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}\{s\} > -\mathrm{Re}\{a\}$
Power-Exponential	$\frac{t^n \exp(-at)u(t)}{n!}$	$\frac{1}{(s+a)^{n+1}}$	$\mathrm{Re}\{s\} > -\mathrm{Re}\{a\}$

Assume that $x_1(t)$ and $x_2(t)$ have Laplace transforms $X_1(s)$ and $X_1(s)$ and R_{x_1} and R_{x_2} are the respective regions of convergence.